Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

go_enrichment_analysis.py

A list of a few thousand gene names does not mean much on its own, so the script asks what those genes are doing as a group. It sends the DEG lists to g:Profiler, which checks whether any biological processes or pathways appear more often in the list than you would expect by chance.

- **enrichment_q1_up.csv, enrichment_q1_down.csv, enrichment_q2_up.csv, enrichment_q2_down.csv, enrichment_overlap.csv (and matching PNGs):** one table and one bar plot for each gene set. Each row is an enriched term (a GO process or a pathway) with how many of my genes fall into it and the corrected p-value. The up and down lists are kept separate on purpose, because pooling them mixes two opposite biological stories and blurs the result. The files are only written when terms are actually found.
- **q1_transcription_factors.csv, q2_transcription_factors.csv:** the transcription factors that appear among the DEGs themselves. This is a simple lookup of which DEGs happen to be TFs, and it is not the same as the TF activity analysis later on, which infers which TFs are active from the behaviour of their targets.
- **tf_families.png:** a bar plot of which families those transcription factors belong to, to see whether any one family dominates.
- **correlation_top50.png, correlation_matrix_top50.csv, correlated_gene_pairs.csv:** how the top 50 genes move together across the samples, as a heatmap, as the underlying matrix of correlation values, and as a list of the most strongly correlated pairs. Genes that rise and fall together are often part of the same process.
- **q3_overlap_detailed.csv, overlap_fc_scatter.png:** the Q3 overlap genes in full, and a scatter plot of their Q1 fold change against their Q2 fold change. The important thing in the scatter is that most points sit in the opposite quadrants, meaning genes that go down with the mutation tend to go up with age. That is early evidence that the mutant is not simply accelerated ageing.